

CHANDRASHEKAR BALA

Cybersecurity Analyst | SOC Analyst | Security Operations | Threat Intelligence

✉ chandrashekar-bala@protonmail.com

☎ +91 9133093202

📍 Hyderabad, Telangana

🌐 in/chandrashekar-bala

🐙 github.com/chandrashekar-bala

🌐 chandrashekar-bala.github.io

PROFESSIONAL SUMMARY

Cybersecurity professional with hands-on independent experience across security operations, threat intelligence, security monitoring, vulnerability management, digital forensics, malware analysis, and network security. Experienced in log analysis, alert triage, IOC identification, threat detection, incident-response investigation workflows, MITRE ATT&CK mapping, OSINT investigations, network traffic analysis, vulnerability validation, and Windows forensic analysis through structured security environments. Strong Linux, Windows, Python, Bash, C, Git, GCC, GDB, and security-tooling background, with practical experience investigating suspicious activity, analyzing attacker behavior, documenting findings and attack paths, and developing repeatable security workflows.

EXPERIENCE

Cybersecurity Researcher — Independent

Jul'24 – Present

Independent Cybersecurity Research & Security Engineering | Self-employed

- Conduct independent cybersecurity research and hands-on security assessments across security operations, threat intelligence, network, web application, wireless, and system security.
- Analyze threat intelligence reports and map observed tactics, techniques, and procedures (TTPs) to the MITRE ATT&CK framework to understand attacker behavior.
- Conduct OSINT investigations involving domains, IP addresses, infrastructure, threat reports, malware, and Windows forensic artifacts.
- Analyze network traffic and packet captures using Wireshark to investigate protocols, communication patterns, suspicious activity, indicators of compromise, and attacker behavior.
- Practice security monitoring workflows including log analysis, alert triage, IOC identification, threat detection, and incident-response investigation in structured security laboratories.
- Perform vulnerability assessment, network reconnaissance, service enumeration, and controlled security testing using Nmap and Linux-based security tooling.
- Conduct web application security testing with Burp Suite and OWASP ZAP, including SQL injection, XSS, authentication, and access-control testing in controlled environments.
- Perform Windows forensic and malware-analysis exercises covering registry, file-system, disk, suspicious-file, and static/dynamic analysis.
- Develop Python and Bash tooling for security research, repeatable analysis, reconnaissance, and basic automation workflows.
- Maintain technical documentation covering findings, indicators, testing methodology, attack paths, and remediation considerations.

Security Researcher — Independent

Jul'26 – Aug'26

Independent Security Research & Security Engineering | Self-employed

- Completed Linux kernel compatibility and wireless-security engineering work involving a Realtek RTL8812BU/RTL8822BU USB Wi-Fi driver.
- Analyzed driver source and kernel compatibility issues and implemented targeted fixes across 13 source files for newer Linux kernel APIs and build requirements.
- Diagnosed compiler, kernel API, and module-build issues using C, Linux kernel headers, GCC, conditional compilation, Git, and Linux kernel build workflows.
- Built and validated the modified kernel module on Kali Linux using real RTL8812BU hardware, confirming monitor mode, packet injection, packet capture, Wireshark, and Aircrack-ng workflows.

PROJECTS

Linux Kernel Compatibility & Wireless Security Engineering

Jul'26 – Aug'26

C | Linux Kernel | RTL8812BU/RTL8822BU | Kali Linux | Git

- Modified an existing Realtek USB Wi-Fi driver to address compatibility issues introduced by newer Linux kernel APIs and build requirements.
- Analyzed kernel-facing C code and implemented targeted compatibility changes across 13 source files while preserving the existing driver architecture and functionality.
- Diagnosed kernel-module build and API compatibility issues using C, Linux kernel headers, GCC, conditional compilation, Git, and Linux kernel build workflows.
- Built, installed, and validated the resulting kernel module on Kali Linux using real RTL8812BU hardware.
- Validated monitor mode, packet injection, packet capture, Wireshark analysis, and Aircrack-ng workflows as part of wireless-security testing.

Web Application Security Assessment Lab

Feb'26 – Present

PortSwigger Web Security Academy | Burp Suite | OWASP ZAP

- Conduct hands-on web application security testing covering SQL injection, cross-site scripting, authentication, access-control, and related OWASP Top 10 vulnerabilities.
- Analyze requests, responses, parameters, sessions, authentication mechanisms, and application behavior to identify vulnerability conditions and attack paths.
- Reproduce findings in controlled labs and document exploitation conditions, security impact, and remediation approaches.

PROJECTS & TECHNICAL RESEARCH

Vulnerability Assessment & Privilege Escalation Lab
VulnHub Virtual Machines / Nmap / Linux / Metasploit

Dec'25 – Present

- Perform reconnaissance, service enumeration, vulnerability assessment, system enumeration, and controlled exploitation across isolated virtual machines.
- Practice Linux privilege escalation and post-exploitation techniques and analyze attack paths from initial reconnaissance through elevated privileges.
- Validate vulnerabilities with controlled exploitation and document findings, exploitation paths, impact, and hardening considerations.

Buffer Overflow Exploit Development Completed
C / Linux / GDB / Exploit Development

- Analyzed stack-based buffer-overflow vulnerabilities and vulnerable program behavior in controlled Linux environments.
- Used GDB and memory-analysis techniques to examine stack layout, control flow, and exploitation conditions and developed proof-of-concept exploitation workflows.

Enterprise Network Penetration Testing Lab Completed
Nmap / Metasploit / Linux / Network Security

- Conducted reconnaissance, service enumeration, vulnerability assessment, and attack-path analysis against a controlled enterprise-style network.
- Investigated exposed services and security weaknesses, validated findings through controlled testing, and documented attack paths and remediation considerations.
- Used Nmap and Metasploit to investigate vulnerable services and understand exploitation paths within an isolated lab environment.

Fake News Detection System — Capstone Project Apr'24
Machine Learning / Python / Natural Language Processing

- Designed and implemented a machine-learning system to classify news articles as legitimate or fake using labeled datasets.
- Performed text preprocessing, feature extraction, model training, and evaluation to assess classification performance.

SKILLS

Security Operations	Security Monitoring, Log Analysis, Alert Triage, Threat Detection, IOC Analysis, Incident Response Workflows, MITRE ATT&CK
Threat Intelligence	OSINT, Threat Intelligence, Threat Report Analysis, TTP Analysis, Infrastructure Analysis, Adversary Research
Vulnerability Management	Vulnerability Assessment, Network Reconnaissance, Service Enumeration, Security Testing, Privilege Escalation, Attack-Path Analysis
Forensics & Malware	Windows Forensics, Registry Analysis, File-System Analysis, Disk Analysis, Malware Analysis, Static/Dynamic Analysis
Network Security	TCP/IP, DNS, DHCP, HTTP/HTTPS, Routing, Switching, Subnetting, CIDR, Network Traffic Analysis, Packet Capture
Web Security	OWASP Top 10, SQL Injection, Cross-Site Scripting, Authentication, Access Control, Burp Suite, OWASP ZAP
Programming & Systems	Python, Bash, C, SQL, Linux, Windows, Git, GCC, GDB, Linux Kernel Headers, Kernel Build Workflows
Security Tooling	Nmap, Wireshark, Burp Suite, OWASP ZAP, Metasploit, Nessus, Aircrack-ng
AI / ML & Frameworks	Machine Learning, NLP, Text Preprocessing, Feature Extraction, Classification, NIST Cybersecurity Framework, OWASP Top 10, MITRE ATT&CK

TOOLS & TECHNICAL STACK

Security Monitoring	Log analysis, alert triage, IOC analysis, network traffic analysis, Windows forensic analysis
Threat Analysis	MITRE ATT&CK, OSINT, TTP analysis, threat-report analysis, infrastructure analysis
Offensive / Validation	Nmap, Burp Suite, OWASP ZAP, Metasploit, Nessus, Aircrack-ng
Programming	Python, Bash, C, SQL
Systems	Linux, Kali Linux, Windows, GCC, GDB, Linux kernel build workflows, Git
Frameworks	MITRE ATT&CK, OWASP Top 10, NIST Cybersecurity Framework

CORE SECURITY CAPABILITIES

SOC / Detection	Security monitoring, log analysis, alert triage, IOC identification, threat detection, incident-response investigation
Threat Research	Threat intelligence, OSINT, threat-report analysis, adversary behavior, TTP analysis, MITRE ATT&CK mapping
Network Analysis	TCP/IP, DNS, DHCP, HTTP/HTTPS, network traffic analysis, packet capture, suspicious-traffic investigation
Endpoint / Forensics	Windows forensic artifacts, registry analysis, file-system analysis, disk analysis, suspicious-file analysis
Vulnerability Analysis	Vulnerability assessment, service enumeration, vulnerability validation, privilege escalation, attack-path analysis
Web Security	OWASP Top 10, HTTP analysis, SQL injection, XSS, authentication, access-control testing, Burp Suite, OWASP ZAP
Automation / Research	Python/Bash scripting, repeatable security workflows, proof-of-concept development, technical documentation

CERTIFICATIONS & PROFESSIONAL TRAINING

Tata — Cybersecurity Analyst Job Simulation	Aug 2026 Forage
Practical simulation covering IAM strategy assessment, solution design, implementation planning, and secure access concepts.	
Google Cybersecurity Professional Certificate	Feb 2026 Coursera
Security, networking, Linux, SQL, Python, SIEM, threat detection, incident response, and SOC workflows. Key Learning: Log analysis, alert triage, vulnerability management, risk assessment, access control, and NIST Cybersecurity Framework.	
Introduction to Blockchain and Web3	May 2024 edX
Blockchain, decentralized systems, Web3 architecture, and security considerations.	
Natural Language Processing Specialization	Apr 2023 Coursera
Text preprocessing, classification, and sequence models using Python.	
Java Full Stack Developer Training	Feb 2023 CipherSchools
Java-based full-stack development and databases.	
Google Cloud Facilitator Program	May 2022 Google
Cloud computing and infrastructure.	

SELECTED TECHNICAL EVIDENCE

- **Security monitoring:** practiced log analysis, alert triage, IOC identification, threat detection, and incident-response investigation workflows in structured security laboratories.
- **Threat research:** conducted OSINT and threat-report analysis, mapped TTPs to MITRE ATT&CK, and investigated attacker behavior and suspicious infrastructure.
- **Network analysis:** analyzed packet captures and network traffic using Wireshark to investigate protocols, communication patterns, suspicious activity, and indicators of compromise.
- **Endpoint and malware analysis:** performed Windows forensic and malware-analysis exercises involving registry, file-system, disk, suspicious-file, and static/dynamic analysis.
- **Vulnerability management:** performed reconnaissance, service enumeration, vulnerability validation, Linux privilege escalation, post-exploitation analysis, and attack-path documentation across isolated virtual machines.
- **Web security:** used Burp Suite and OWASP ZAP to analyze HTTP requests/responses and test SQL injection, XSS, authentication, and access-control vulnerabilities in controlled environments.
- **Security automation:** developed Python/Bash scripts for reconnaissance, repeatable testing, proof-of-concept development, and basic automation.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering	2024
Relevant Coursework: Network Security, Ethical Hacking, Penetration Testing, Machine Learning, Automation	Jalandhar, Punjab
Intermediate (MPC), Telangana Board of Intermediate Education	2020
Mathematics, Physics, Chemistry	Warangal, Telangana

TECHNICAL FOCUS

Security Operations	Security monitoring, log analysis, alert triage, IOC analysis, threat detection, incident-response investigation workflows
Threat Intelligence	OSINT, threat intelligence, threat-report analysis, TTP analysis, adversary behavior, MITRE ATT&CK mapping
Network / Endpoint	Network traffic analysis, packet capture, Windows forensics, registry/file-system analysis, malware-analysis exercises
Vulnerability Management	Vulnerability assessment, service enumeration, validation, privilege escalation, post-exploitation, attack-path analysis

ADDITIONAL TRAINING

- **Java Full Stack Developer Training** — Feb 2023 | CipherSchools — Java-based full-stack development and databases.
- **Google Cloud Facilitator Program** — May 2022 | Google — cloud computing and infrastructure.